Replication of the creative self-efficacy measurement models in Abbott (2010)

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Abstract

Does the revised measurement model for the Creative Thinking Self-Efficacy inventory (CTSE II) and the Creative Performance Self-Efficacy inventory (CPSE II) reproduce the sample correlations, pass the printed fit cuts, and fit more parsimoniously than the rival models? Abbott (2010) answers yes. The refined model has $\chi^2(178) = 295.571$, CFI = .954, RMSEA = .046, and SRMR = .057, and a lower Akaike information criterion than the rival models. CTSE and CPSE correlate with openness to experience and with Beghetto's three-item creative self-efficacy measure. A general factor does not improve the model. We rebuilt the covariance matrix from the printed correlation matrix and refit the Appendix J models by maximum likelihood. Every model has the printed degrees of freedom. The standardized root mean square residual matches to about one thousandth. The implied-structure chi-square matches the maximum-likelihood value implied by the printed scaled chi-square and its scaling factor, 715.13 against 714.98. The revised model remains the most parsimonious, and its Akaike gap against the model that uses every indicator is 234.12 against the printed 235.09. The robust chi-square, comparative fit index, and root mean square error of approximation of the revised model are not recovered from the printed matrix. The interview phase is not a covariance model and is not refit.

1 The question, and the headline

This section answers whether the printed measurement model is the one in Appendix J, and whether its fit statistics and its rank against the rival models come back from the printed correlation matrix. The degrees of freedom and the parsimony order do. The standardized residual index does, to about one thousandth. The robust chi-square and the two indices that are computed from it do not, on this matrix.

Abbott (2010), https://digitalcommons.unl.edu/cehsdiss/68/, asks how creative self-efficacy is structured. Creative thinking self-efficacy has four factors: fluency, flexibility, elaboration, and originality. Creative performance self-efficacy has three: domain, field, and personality. The quantitative phase fits that model, a general-factor restriction, and four rival models, by confirmatory factor analysis. The program named on the Appendix J listing is Mplus 5.2, with the robust maximum likelihood estimator (MLR). The sample that began the instrument is 308 people. The correlation matrix is printed at $n = 296$. The fourth research question is about interviews. Those transcripts are not a covariance model, and this note does not refit them.

The original work is the inventory, the design, and the printed tables. The calculations below are ours: the covariance matrix read from Table 3.2, the maximum-likelihood fits, and the comparison. Figure 1 is the initial-model line on printed page 129.

The Revised Model. The Revised model was tested. After loadings of the four factors indicating CTSE (elaboration, flexibility, fluency, and orientation) and the three factors indicating CPSE (domain, field, and personality) were inputted, the model does not initially pass Hu and Bentler's (1999) criteria, $\chi^2(181) = 350.161$, $p < .01$, Scaling = 1.456, CFI = .934, RMSEA = .055, SRMR = .061, and AIC = 52830.056 . Several empirically justified latent correlations between three pairs of manifest indicators were

Figure 1: Printed page 129. The initial model and the word "orientation" in the sentence that names the four thinking factors.

2 What we replicated

This section answers which printed claims agree with a direct fit to Table 3.2. Table 1 is that comparison. "Match" means the same degrees of freedom, the same rounded index, or the same ordering. "Close" means a gap on the order of the third printed decimal, or a chi-square gap of a few points on a difference. "Partial" means one piece of the claim holds and another does not. "No match" means a different number after the checks in Section 3.

The degrees of freedom fix the specification. For 21 revised indicators the initial model has 231 sample moments and 50 free parameters, so the degrees of freedom are 181. Three residual covariances bring that to 178. Freeing the two second-order marker loadings and fixing the two second-order variances and their covariance at 1 changes the count by one, to 179. Those are the printed degrees of freedom. The same count holds for the six larger models, including the all-indicator model at 536 and the held model at 311.

The standardized root mean square residual is computed from the correlation residuals of the maximum-likelihood point estimates. It does not use the robust scaling factor. The initial model gives .0612 against the printed .061. The all-indicator model gives .0621 against .062. The held model gives .0599 against .060. The refined model gives .0578 against .057. Figure 2 is the refined line.

correlations between these three empirically and justified pairs of manifest indicators. This refined Revised model passed Hu and Bentler's (1999) criteria, $\chi^2(178) = 295.571$, $p < .01$, Scaling = 1.463, CFI = .954, RMSEA = .046, SRMR = .057, and AIC = 52758.669. A Satorra (2000; Satorra & Bentler, 2001) scaled χ^2 difference test indicated a significant improvement in performance between these models, Satorra-Bentler $\chi^2(3) = 52.457$, p < .

Figure 2: Printed page 130. The refined model and the scaled difference of 52.457 on 3 degrees of freedom.

The scaled difference uses the printed chi-squares and the printed scaling factors. With T the

scaled chi-square, d the degrees of freedom, and c the scaling factor,

$$c_d = \frac{d_0 c_0 - d_1 c_1}{d_0 - d_1}, \qquad \frac{T_0 - T_1}{c_d}.$$

For the initial model against the refined model this is 52.457, the printed value. The function is `simulation/fit_models.R`. The maximum-likelihood chi-square difference on our fits is 75.08 on 3 degrees of freedom. The difference of the two unscaled products, $350.161 \times 1.456 - 295.571 \times 1.463$, is 77.41. The two differences sit 2.3 apart.

Mplus prints a scaled chi-square under MLR and a scaling factor such that the product is the ordinary maximum-likelihood chi-square. For the implied-structure model that product is 714.98, and our chi-square is 715.13. For the implied-indicators model the product is 780.21 and our chi-square is 778.45. For the revised model the product is 432.42 and our chi-square is 450.69. The comparative fit index and the root mean square error of approximation printed for the revised model are computed from the scaled chi-square. Our maximum-likelihood versions are .939 and .072. They do not pass the cuts CFI $> .95$ and RMSEA $< .06$. The printed robust indices do pass those cuts. The standardized residual index, which does not use that scaling, passes SRMR $< .09$ on both sides.

The Akaike information criterion is on different numeric levels because it depends on the sample size and on $\log |\Sigma|$, and the printed standard deviations are rounded to one decimal. On the models that use the same observed variables, the order is the printed order. The revised model with saturated correlates has the lowest value, then the implied-structure model, the implied-indicators model, the all-indicator model, and the held model. The gap against the all-indicator model is 234.12 against 235.09. The gap against the implied-structure model is 34.38 against the printed 64.47. That second gap is the chi-square gap between those two models, shifted by two points for the extra parameter. Our chi-square gap is about 36, and the gap of the unscaled printed chi-squares is about 67.

Beghetto's measure correlates with creative thinking self-efficacy at .529 against .53, and with creative performance self-efficacy at .685 against .69. Openness correlates with those two factors at .435 against .45 and .523 against .53. Across the 55 off-diagonal entries in Table 4.1, the mean absolute gap is 0.0065 and one entry exceeds 0.02: openness with flexibility, .390 against .41. The file is `simulation/out/table41_gaps.csv`.

The Kaiser–Meyer–Olkin index on all 37 columns of Table 3.2 is 0.925, which rounds to the printed .93. The index on the creative-thinking and creative-performance items alone is 0.935. The function is `simulation/fit_models.R`.

The first-order composite reliabilities use the square of the sum of the standardized loadings in the numerator. On the refined model that quantity is 0.596 for fluency, 0.910 for domain, 0.842 for field, and 0.890 for personality, against the Table 4.2 entries 0.60, 0.91, 0.84, and 0.89. Elaboration is 0.771 against 0.78. Flexibility is 0.882 against 0.93, and originality is 0.935 against 0.88. The ratio written on printed page 110, with the sum of squared loadings in the numerator, gives fluency 0.356 and does not match those cells. Both columns are in `simulation/out/composite_validity.csv`.

3 What gave trouble, and how it was resolved

This section answers where a first reading disagreed with a printed number, and which change removed the disagreement. The disagreements that remain are in Section 7.

The sample size and the likelihood were the first check. Table 3.2 prints $n = 296$. Table 4.1 prints $n = 308$. The text also says 297 people completed every item. A grid of

296, 297, and 308, each with the Wishart likelihood and with the normal likelihood, is in simulation/out/n_likelihood_grid.csv. The Wishart fit at 296 is the closest of those six to 350.161×1.456, the unscaled target for the initial model. The remaining gap there is 15.9 chi-square points. Changing the sample size multiplies the chi-square. It does not remove that gap. The same Wishart fit at 296 does land on the implied-structure target, so the multiplier is not a single global error.

The robust scaling was the second check. Dividing our refined chi-square by the printed scaling factor 1.463 produces 308.1, against the printed scaled value 295.571. The comparative fit index and the root mean square error of approximation move with that scaled value, and with the baseline model's own scaling factor, which is not printed. We therefore compare the ordinary maximum-likelihood chi-square with the product of the printed scaled chi-square and the printed scaling factor, and we compare the standardized residual index directly.

The optimizer was the third check. lavaan [2] with `nlminb` and with BFGS agree to about 0.07 chi-square points on the initial, refined, general-factor, and span models. OpenMx [3] with SLSQP and with CSOLNP agree with each other to 10^{-5}. OpenMx multiplies the fit function by N. lavaan's Wishart likelihood multiplies it by $N - 1$. Rescaling the lavaan initial chi-square by 296/295 gives 527.56. The OpenMx value is 527.52. The refined and general-factor models show the same relationship. The file is simulation/out/openmx_fits.csv. The optimizer is not the source of the gap with the printed robust chi-square.

The saturated-correlate models needed a separate start. lavaan's default starts make the initial model-implied matrix fail to be positive definite once unused items are given free covariances with every other item. Starting those covariances at zero, and starting residual variances from the sample variances, lets the models converge. The iteration limit in simulation/fit_models.R is 3000. The revised model with saturated correlates then has chi-square 678.87 and 326 degrees of freedom, against 678.86 and 326 for the same focal model without the extra items. Adding a saturated item does not change the degrees of freedom or, here, the chi-square. That is the pattern in the printed pair 514.150 and 514.108 as well. lavaan warns that the residual covariance matrix is not positive definite for the four models that carry those extra covariances. The implied-structure model carries that warning and still matches its unscaled target, so the warning alone does not identify the revised model's chi-square gap.

The general-factor difference test was checked as algebra before it was treated as a discrepancy. The printed degrees of freedom are 179 and 178, so a nested comparison has 1 degree of freedom. The code frees the two marker loadings and fixes three second-order moments, which is the same net change of one parameter. The algebra that returns 52.457 for the three residual correlations returns +4.02 for this comparison. The printed line is $\chi^2(2) = -21.803$. Figure 3 is that line. Our maximum-likelihood difference is +40.73 on 1 degree of freedom, against the unscaled printed difference 42.94. The general-factor model fits worse on both accounts. The negative two-degree-of-freedom figure is the cell that remains.

The parenthetical on that page says the general-factor comparative fit index is .054 and the criterion is .05. The fit line two paragraphs above prints CFI = .945. The criterion stated for that index earlier in the dissertation is .95. The comparison in Table 1 uses the fit line.

The word on printed page 129 for the fourth thinking factor is "orientation." Appendix J names the factor with the originality items `CTORIG1` through `CTORIG4`. The model we fit uses those items. Figure 1 shows both the word and the fit line.

rival alternative model, however, did not pass Hu and Bentler's (1999) criteria, $\chi^2(179) = 319.462$, $p < .01$, Scaling = 1.488, CFI = .945, RMSEA = .050, SRMR = .055, and AIC = 52799.519. A visualization of this rival model appears as Figure 4.3. A Satorra (2000; Satorra & Bentler, 2001) scaled χ^2 difference test of these models produced a negative result, Satorra-Bentler χ^2 (2) = -21.803, which means that the difference test is not interpretable (B.O. Muthén, 2006).

While the General Factor model of creative self-efficacy did not pass Hu and Bentler's criteria, the small difference between the general factor's model CFI (.054) and Hu and Bentler's criterion for CFI (.05) made it reasonable to examine this rival model

Figure 3: Printed page 131. The general-factor fit, the negative difference, and the parenthetical comparative fit index.

4 What we did ourselves, and what we took from others

This section answers which pieces are ours and which pieces are taken from the dissertation. The dissertation contributes the inventories, the factor structure, the rival models, and the printed tables. We contribute the covariance matrix read from Table 3.2, the maximum-likelihood fits, the OpenMx check, and the comparison.

The three residual covariances are the ones named in the text and written in Appendix J: elaboration items 4 and 5, domain items 1 and 4, and personality item 2 with field item 2. On the initial model, those three pairs are among the larger modification indices for residual covariances. The largest modification index is a factor residual covariance, which the dissertation does not free. We did not add it.

No number in the comparison is taken from a later paper. The composite-reliability formula with the square of the sum of the loadings is the usual one. The dissertation's page 110 writes the sum of squared loadings in the numerator. Both are computed in the output file named above. Table 4.2's fluency, domain, field, and personality entries agree with the usual formula.

5 Assumptions

Each subsection is a choice the dissertation does not settle and that the fits require.

5.1 The printed matrix is one complete-data covariance matrix

Table 3.2 prints a correlation, a mean, and a standard deviation for each of 37 columns, at $n = 296$. Eleven of the 308 people who started the instrument left at least one item blank, and 297 people completed every item. The dissertation estimates the models by full-information maximum likelihood on the raw file. We treat the printed correlations and standard deviations as the covariance

matrix of one complete sample of size 296. The standard deviations enter only the scale of the covariance matrix. A confirmatory factor model with a marker loading fixed at 1 is invariant to rescaling each observed variable, so the chi-square does not depend on those standard deviations. It does depend on the correlations. The correlations are printed to two decimals. Figure 4 shows the fluency item whose standard deviation is 52.3, and a minus sign on a later row.

26: CTFu3	22.5	10.5	.15	.14	.12	.28	.08	.19	.16	.31	.21	.19	.19	.29	.22	.16	.23	.15	.28	.11	.10	.28	.32	.31	.28	.26	.36	–
27: CTFu4	74.8	52.3	.23	.17	.12	.28	.21	.26	.23	.10	.24	.22	.20	.24	.21	.18	.24	.20	.27	.21	.18	.32	.34	.32	.30	.32	.35	.20	–
28: CTO1	63.8	24.0	.48	.47	.34	.56	.34	.42	.41	.24	.59	.58	.53	.47	.51	.45	.60	.42	.46	.39	.40	.52	.56	.42	.42	.49	.51	.28	.2(
29: CTO2	70.5	22.2	.50	.46	.32	.47	.34	.41	.40	.26	.51	.60	.53	.53	.49	.45	.60	.46	.53	.42	.40	.61	.58	.44	.55	.51	.51	.27	.2
30: CTO3	67.2	21.7	.46	.40	.30	.51	.39	.45	.46	.22	.52	.59	.49	.50	.44	.51	.57	.48	.59	.39	.40	.64	.62	.49	.55	.49	.54	.24	.2
31: CTO4	64.6	22.8	.51	.48	.40	.54	.36	.43	.45	.24	.60	.64	.60	.57	.58	.53	.67	.48	.54	.45	.42	.59	.63	.48	.53	.54	.56	.30	.2
32: Open1	5.5	1.4	.29	.40	.70	.19	.17	.22	.21	-.02	.23	.31	.25	.19	.27	.21	.25	.24	.20	.19	.26	.18	.20	.21	.22	.26	.14	.00	.1
33: Open2	3.1	1.6	-.22	-.21	-.22	-.15	-.11	-.14	-.12	-.11	-.11	-.15	-.13	-.26	-.20	-.23	-.23	-.18	-.13	-.12	-.13	-.17	-.13	-.15	-.17	-.21	-.08	-.05	-.1
34: Open3	3.3	1.5	-.28	-.30	-.24	-.19	-.15	-.18	-.24	.02	-.16	-.22	-.13	-.20	-.15	-.24	-.21	-.19	-.23	-.14	-.17	-.25	-.24	-.22	-.20	-.16	-.20	-.07	-.1
35: Open4	2.2	1.5	.32	.31	.61	.26	.22	.31	.29	.07	.15	.29	.25	.25	.22	.20	.25	.26	.25	.27	.23	.21	.19	.19	.22	.23	.16	.07	.1

Figure 4: Printed page 95. The row for fluency item 4, with standard deviation 52.3, and minus signs in the rows below it.

5.2 Row labels map onto the Appendix J names in printed order

The table's note gives the block order: Beghetto's three items, four domain items, four field items, four personality items, elaboration, flexibility, fluency, originality, four openness items, sex, and age. Elaboration begins at item 2, so `CTELAB1` is not in the table. The tenth row is labeled `CPFI` and sits between field items 2 and 4. We read it as field item 3, `CPFIEL3`. Both the revised model and the implied-structure model use that item as an indicator. The implied-structure chi-square matches its unscaled target under this reading.

5.3 The Wishart likelihood at $n = 296$

lavaan can multiply the fit function by $N - 1$ or by N. Mplus 5.2's ordinary maximum-likelihood chi-square is the fit function times a sample-size factor as well. We use the Wishart choice, $N - 1$, at the sample size printed on Table 3.2. Section 3 records the grid. Latent means are fixed at zero and observed intercepts are free, so the mean structure is saturated and does not add to the chi-square.

5.4 Marker identification, with the general-factor loadings free

Appendix J fixes the first loading of each first-order factor at 1. In the general-factor model the first loading of each second-order factor is free, and the two factor variances and their covariance are fixed at 1. lavaan will put a loading of 1 back on the first indicator if the syntax names that indicator twice. The syntax uses a single free parameter on each of those two loadings. The estimates are free in `simulation/out/general_second_order_loadings.csv`.

5.5 Openness items 2, 3, and 4 keep the printed signs

Those three columns correlate negatively with the creative self-efficacy items. Openness item 1 correlates positively with them. We do not reverse the three columns before the fit. Their loadings

start negative so that the first model-implied matrix can be positive definite. The signs of the loadings are estimated, not fixed.

6 Ambiguities that were resolved

This section answers which printed details were ambiguous and which file now settles them.

The tenth row of Table 3.2 is labeled `CPFI`. The resolution is the order in the table note together with the implied-structure chi-square. The script that reads the text layer is `simulation/parse_table32.py`, and the matrix is `simulation/table32.csv`. Regenerating the matrix from the PDF is `python3 formalization/abbott-ctse/simulation/parse_table32.py` with the PDF path. The committed matrix matches that reading.

The sentence on printed page 110 writes composite validity as $\sum \lambda^2/(\sum \lambda^2 + \sum \varepsilon)$. The Table 4.2 entries for fluency, domain, field, and personality match the square of the sum of the standardized loadings instead. The output file records both. The command is `Rscript fit_models.R out` from `formalization/abbott-ctse/simulation`.

The saturated-correlate syntax is the block of `WITH` statements in Appendix J. Each unused item is given a free covariance with every other observed variable, once. The function is `simulation/fit_models.R`. After the starts in Section 3, those models converge, and their degrees of freedom match the printed degrees of freedom.

7 What remains unresolved

This section answers which printed numbers are still apart from the fits, and which checks did not close the gap.

The refined model's ordinary chi-square is 450.69 against the unscaled target 432.42. The initial model is 525.77 against 509.83. The span model is 678.86 against 648.34. The same matrix produces 715.13 against 714.98 for the implied-structure model, and 778.45 against 780.21 for the implied-indicators model. The gap is therefore not a single multiplier on every model. Full-information maximum likelihood on the raw file, with eleven partially observed people, is the check that would use a different matrix for each pattern of missing items. The file named in Appendix J, `CohortsTwo.dat`, is the file that would settle it. Two-decimal rounding of the correlations is a second open contributor. A perturbation of each correlation by up to half a hundredth did not return a stable band: the perturbed matrices did not converge under the same starts. That attempt is not used as a tolerance.

The comparative fit index and the root mean square error of approximation of the refined model remain the robust versions. Recomputing them requires the baseline model's scaling factor as well as the focal one. The printed focal scaling factor is 1.463. The baseline factor is not printed.

The negative difference is the other cell. The inputs are the printed chi-squares 319.462 and 295.571, the scaling factors 1.488 and 1.463, and the degrees of freedom 179 and 178. The algebra that reproduces 52.457 gives +4.02 on 1 degree of freedom. The printed line is −21.803 on 2 degrees of freedom. The models are not a pair in which one set of free parameters is a subset of the other in a single parameterization: the general-factor model frees the marker loadings and fixes the second-order moments. A raw-data run in Mplus 5.2, with the scaling factors and the difference test written to the output, is the check.

The fraction of correlations greater than .30 is printed as .66. On all 37 columns the fraction of absolute correlations above .30 is .544. On the creative-thinking and creative-performance items it

is .817. Within the factor blocks it is .843. None of those readings is .66. The Kaiser–Meyer–Olkin index from the same matrix does round to .93.

The Akaike gap against the implied-structure model is 34.38 against 64.47. That gap moves with the revised model's chi-square. It is the same open question as the chi-square level, not a separate one.

Smaller differences, kept out of the verdicts above, are these. The refined standardized residual rounds to .058 against the printed .057. The general-factor residual rounds to .056 against .055. The implied-structure residual rounds to .050 against .049. The span residual rounds to .059 against .058. Beghetto's correlation with creative performance self-efficacy rounds to .68 or .69 depending on the rounding rule; the printed value is .69 and ours is .685. These are the third-decimal gaps. They are not, by themselves, a failure of the degrees of freedom or of the parsimony order. Figure 5 is the correlation table.

Latent Factor	SD	1	2	3	4	5	6	7	8	9	10	11
1: BCSE	0.93	–										
2: CPDOMA	14.6	0.50	–									
3: CPFIEL	16.51	0.64	0.68	–								
4: CPPERS	16.46	0.58	0.62	0.80	–							
5: CPSE	10.57	0.69	0.72	0.94	0.85	–						
6: CTELAB	17.65	0.42	0.48	0.63	0.57	0.67	–					
7: CTFLEX	18.55	0.48	0.56	0.72	0.65	0.77	0.70	–				
8: CTFLUE	15.04	0.47	0.55	0.71	0.64	0.75	0.68	0.78	–			
9: CTORIG	20.03	0.46	0.53	0.69	0.63	0.74	0.67	0.77	0.75	–		
10: CTSE	13.16	0.53	0.62	0.81	0.73	0.86	0.78	0.89	0.88	0.86	–	
11: OPEN	0.88	0.65	0.38	0.49	0.45	0.53	0.35	0.41	0.40	0.39	0.45	–

Note. The sample size is n = 308. All correlations (r > 0.10) significant at the p < .05 level. All correlations

Figure 5: Printed page 135. Latent correlations for the revised model with Beghetto's measure and openness.

8 Code and source availability

This section answers where the dissertation and the analysis code were sought. The dated rows are the ledger in `simulation/out/code_search.md`.

Place	Date	Result
UNL PDF via Wayback `20261004155053`	2026-10-29	Obtained. sha256 `a3c381d6...`
Direct curl of the UNL download URL	2026-10-29	HTTP 403, bot check
ProQuest docview `305217740`	2026-10-29	HTML page. PDF bytes not saved
Ledger copy of the same dissertation	2026-10-29	Different sha256, file not in this fit
Appendix J path `CohortsTwo.dat`	2026-10-29	Local path only. Not obtained
GitHub, GitLab, OSF, Zenodo	2026-10-29	No analysis repository and no raw file
Mplus 5.2	2026-10-29	Named in Appendix J. Not installed

The PDF used for the crops and for the matrix has sha256 `a3c381d6f6295e917806834364b7f5 5fc35684b3db53174a53c3564feab4d70a`. It is a 283-page file whose producer line is Prince 15.4.1 and whose creation date is 30 August 2026. The literature ledger records a different hash, `d39a6 96d20d6df6d9467a8906e67c40f336c3856ac74637f8db6629ca9fcc9d2`, for a file of the same page count. The fits and the crops in this note are from the file we obtained.

R [1] is version 4.3.3. It is the program named, in the 2009 citation, as the general tool for the quantitative work. lavaan [2] is version 0.6.17, from https://CRAN.R-project.org/package=lavaan. It fits the mean and covariance model that Appendix J specifies, because Mplus 5.2 is not installed here. OpenMx [3] is version 2.21.11, from https://CRAN.R-project.org/package=OpenMx. It is the second optimizer. The versions are the `sessionInfo` record in simulation/out/sessionInfo.txt.

9 Next steps

The raw file `CohortsTwo.dat`, or an Mplus 5.2 output that prints the unscaled chi-square, the baseline scaling factor, and the difference test, is the check that separates the revised model's chi-square level from the robust scaling and from full-information maximum likelihood. The same file would recompute the fraction of correlations above .30. The ProQuest file whose hash is recorded in the literature ledger would show whether any printed entry in Table 3.2 differs from the file used here. The interview phase remains outside this comparison. It would be a reading of the transcripts, not a refit of Table 3.2.

10 Artifacts

- Matrix. File simulation/table32.csv. Command: `python3` simulation/parse_table32.py with the PDF.
- Fits. File simulation/out/comparison.csv, simulation/out/lavaan_fits.csv, simulation/out/openmx_fits.csv. Command, from `formalization/abbott-ctse/simulation`: `Rscript` simulation/fit_models.R `out`.
- Sample-size grid. File simulation/out/n_likelihood_grid.csv. Same command.
- Scaled-difference algebra. File simulation/out/satorra_printed.csv. Function simulation/fit_models.R.
- Latent correlations. File simulation/out/table41_gaps.csv.
- Composite reliability. File simulation/out/composite_validity.csv.

- Matrix indices. File simulation/out/matrix_checks.csv. Function simulation/fit_models.R.
- Rounding draws. File simulation/out/rounding_band.csv. Three draws, seed 20100501. Each chi-square is missing because the perturbed matrix did not converge.
- OpenMx. Function simulation/fit_models.R. File simulation/out/openmx_fits.csv.
- Availability ledger. File simulation/out/code_search.md.
- Page crops. Directory paper/figures. Command, from the repository root: `python3` paper/figures/crop_pages.py.

The dissertation PDF is local and is not in git.

References

[1] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024. https://www.R-project.org/.

[2] Yves Rosseel. lavaan: An R package for structural equation modeling. *Journal of Statistical Software* 48(2) (2012), 1–36. 10.18637/jss.v048.i02. Package page: https://CRAN.R-project.org/package=lavaan.

[3] Michael C. Neale, Michael D. Hunter, Joshua N. Pritikin, Mahsa Zahery, Timothy R. Brick, Robert M. Kirkpatrick, Ryne Estabrook, Timothy C. Bates, Hermine H. Maes, and Steven M. Boker. OpenMx 2.0: Extended structural equation and statistical modeling. *Psychometrika* 81(2) (2016), 535–549. 10.1007/s11336-014-9435-8. Package page: https://CRAN.R-project.org/package=OpenMx.

Claim	Published	Ours	Verdict
Degrees of freedom, nine Appendix J models	The nine values in the caption	The same nine numbers	match
Refined model, SRMR	.057	.0578	close
Refined model, CFI and RMSEA	.954 and .046	.939 and .072, maximum likelihood	partial
Scaled difference for three residual correlations	$\chi^2(3) = 52.457$	52.457 from the printed inputs	match
Maximum-likelihood difference for those correlations	Unscaled difference 77.41	75.08 on 3 df	close
General factor is the worse model	CFI .945, higher AIC	CFI .930, higher chi-square	match
Negative scaled difference	$\chi^2(2) = -21.803$	+4.02 on 1 df from the same algebra	no match
Parsimony order	Revised, then implied structure, implied indicators, all, held	The same order	match
AIC gap, revised against all indicators	235.09	234.12	match
AIC gap, revised against implied structure	64.47	34.38	no match
Implied-structure chi-square, unscaled	$526.884 \times 1.357 = 714.98$	715.13	match
BCSE with CTSE and with CPSE	.53 and .69	.529 and .685	close
Openness with CTSE and with CPSE	.45 and .53	.435 and .523	close
Kaiser–Meyer–Olkin index	.93	.925 on all 37 columns	match

Table 1: Printed claims against the fits in `simulation/out/comparison.csv`. BCSE is Beghetto's creative self-efficacy measure. AIC is the Akaike information criterion. The nine degrees of freedom belong, in order, to the initial model, the refined model, the general-factor model, the span model, the revised model with saturated correlates, the all-indicator model, the held model, the implied-structure model, and the implied-indicators model.